Experiment No. 26 – Reverse Video Frames Using OpenCV

1. Aim

To implement a function to reverse the frames of a video and create a video in reverse mode using OpenCV.

2. Program

```
import cv2

from google.colab import files

from IPython.display import Video, display


# Upload video

uploaded = files.upload()

filename = list(uploaded.keys())[0]


# Open video

cap = cv2.VideoCapture(filename)


# Get video properties

fps = cap.get(cv2.CAP_PROP_FPS)

width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))

height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))


# Store video frames

frames = []


while True:
```

```
    ret, frame = cap.read()

    if not ret:
        break

    frames.append(frame)

cap.release()

# Reverse the frames
frames.reverse()

# Create reverse video
output_file = "/content/reverse_video.mp4"

fourcc = cv2.VideoWriter_fourcc(*'mp4v')
writer = cv2.VideoWriter(
    output_file, fourcc, fps,
    (width, height)
)

for frame in frames:
    writer.write(frame)

writer.release()
```

```
print("Reverse video created successfully!")

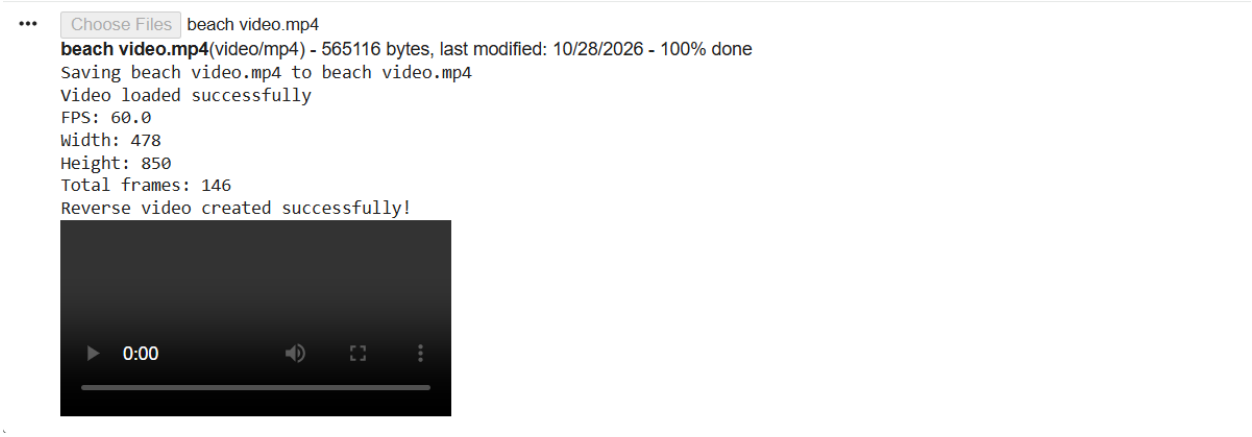

# Display reverse video

display(Video(output_file, embed=True))
```

3. Expected Output

The output video plays the frames in reverse order, making the video appear to move backwards.

4. Output



The input video frames are successfully reversed, and a new reverse video is generated.

5. Result

Thus, the video frame reversal technique was successfully implemented using Python and OpenCV.